

Surface Rings: The “Different” Particles

A Short Front-End and Call for Feedback

Hono Salval*

November 6, 2025

Abstract

We present a particle-like framing of Surface Rings (S-rings), reserving *surfeon* as the technical name for the resulting boundary excitation. Starting from the split-skew representation theorem for passive, causal boundary laws, we show how S-rings support lossless spectral bands (“rings”) and how a surfeon emerges on such bands; how “mass”, charges and couplings derive from *positive spectral measures*; and how interconnection rules supply effective interactions. This front-end summarizes the canon, algebra, and construction recipe, and outlines calibrated checks (GR optics/timing, atomic integrals, GW backgrounds) to invite early attribution and feedback. A rigorous companion note provides proofs and extended examples.

1 Introduction: From Interfaces to Excitations

S-rings place boundary physics at center stage: given an oriented separator S , fold field traces into “difference” and “consensus” slots, constrain the law by causality, passivity, and surface symmetry, and solve the resulting spectral problem. The representation theorem forces a two-slot split-skew algebra with a positive (Herglotz/Stieltjes) part and a lossless, orientation-odd skew. In this lens, particle-like behavior is not postulated in the bulk but *selected* at the boundary: lossless bands in (ω, K) define compact, robust excitations.

We therefore adopt the following naming: the *canon* is **Surface Rings** (S-rings); a *surfeon* is a technical term for the particle-like excitation carried by an S-ring band. This keeps the continuous derivation visible while giving a crisp, symmetry-aware bridge to modern particle language.

2 Split-Skew Canon and Representation

Let S be an oriented surface. Fold value/flux traces into jump/average slots $\Psi = \begin{pmatrix} [q] \\ \{q\} \end{pmatrix}$ and $\Theta = \begin{pmatrix} [p] \\ \{p\} \end{pmatrix}$ with the boundary power pairing. Under causality, passivity, tangential stationarity and local isotropy, each polarization channel admits the unique linear form

$$\Theta(\omega) = [Z(\omega, K) I + s(K) J] \Psi(\omega), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (1)$$

with $K := \sqrt{-\Delta_S}$. The Hermitian block Z is *Herglotz* in $i\omega$ and *Stieltjes* in K^2 (positive spectral measures); the skew block $s(K) J$ is *lossless* and *orientation-odd*. This is the representation theorem proved in the companion note.

*Companion formal note: “Surface Rings: Formal Definitions, Characterization, and Cross-Domain Validations” (see repository).

Algebra (“ J -complex” structure). Per channel, operators $a(K)I + b(K)J$ form a commutative $*$ -algebra isomorphic to functions of K^2 tensored with $\mathbb{C}$ (map $a + ib \mapsto aI + bJ$), with $J^2 = -I$ and the orientation automorphism $b \mapsto -b$. Physical kernels live in a positive cone (Herglotz/Stieltjes + lossless skew) preserved by passive interconnections (Redheffer/star products).

Automodes (lossless bands). Define *automodes* by vanishing dissipative power: $\ker \Re Z(\omega, K) \neq \{0\}$. Under isotropy, their loci organize as rings in (ω, K) per channel (S-rings). These are the building blocks of surfeons.

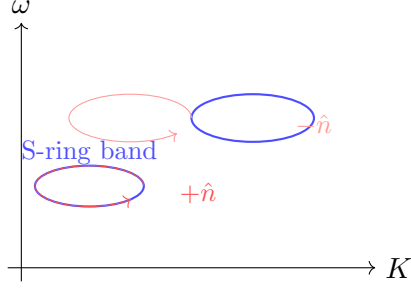
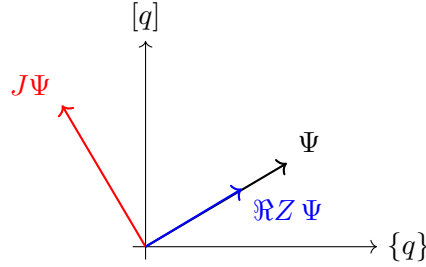


Figure. Automodes organize as closed bands (“rings”) in the (ω, K) -plane. The skew $s(K)$ sets a handed circulation that flips under normal reversal ($\hat{n} \mapsto -\hat{n}$).

Inset (split-skew action).



The Hermitian part $\Re Z$ scales slots identically; the skew J rotates jump/average losslessly; $s(K)$ sets handedness.

3 The Surfeon as “Partition” Particle

Definition (effective excitation). A *surfeon* is a lossless SR automode band supported on S , specified by:

- **Spectral core** $\mathcal{R} = \{(\omega, K) : \ker \Re Z \neq \{0\}\}$ (ring/band);
- **Skew chirality** $s(K)$ (orientation-odd coupling that flips under $\hat{n} \mapsto -\hat{n}$);
- **Positive measures** $(d\mu, d\nu)$ defining Z (dynamics/dispersion).

These data are minimal, symmetry-driven, and independent of a pre-assumed bulk metric. We emphasize: *S-rings* denote the canon; *surfeon* denotes the excitation carried by an S-ring band.

Effective “quantum numbers” (from measures and channels). Surfeon invariants arise from the SR canon:

- **Polarization/spin:** channel labels (scalar/TE/TM/...); chirality via $s(K)$.

- **Mass/threshold:** a UV onset in the Stieltjes measure (support above u_0) induces Yukawa-like decay; the scale u_0 acts as an effective mass via $\ell \sim u_0^{-1/2}$.
- **Charges/couplings:** sector weights (coefficients multiplying shared spectral measures) encode how a surfeon sources/feels other fields; positivity preserves passivity.

Interactions and composition. Passive interconnections glue separators and bulk via Schur complements; at the boundary level this is a star-product on transfer/scattering operators that preserves the split-skew form and positivity. Multi-surfeon composites inherit additive measures and chirality rules.

Construction recipe (knee-free).

1. *Fix channels and symmetry.* Choose the polarization basis and enforce the split-skew constraints.
2. *Choose positive measures.* Specify Z via Herglotz/Stieltjes measures (IR density + optional UV thresholded density); do not insert ad-hoc knees.
3. *Calibrate few IR constants.* Fit small- K coefficients to GR optics/timing (PPN γ , limb deflection, Shapiro profiles).
4. *Freeze and predict.* With IR frozen, compute parameter-sparse overlays (atomic contrasts, GW anisotropy shapes, parity-odd residues); introduce a UV density only if demanded by a cross-check (e.g., Lamb-scale suppression), still as a positive measure.

4 Particles as Surface Rings

We keep the canon continuous and knee-free: $Z(\omega, K)$ is specified by *positive spectral measures* (Herglotz/Stieltjes), and $s(K)$ is lossless and orientation-odd. Surfeons are quantized modes on S-ring bands; S-chains (graphs of rings) produce composite spectra.

Light surfeon chain (lepton-like). A single ring or lightly-coupled dimer with dispersion $\varepsilon_0 + v_K K + \frac{1}{2}\kappa_K K^2$ yields a light edge mode; the inertial scale follows from ε_0 and inter-ring couplings. Sector weights multiply the *same* kernel $\Xi(K) := \Re Z(0, K)$.

Photon-like edge mode. In EM channels, TE/TM surfeons carry two polarizations; a tiny $s^{(1)}(K)$ gives an orientation-odd birefringence without spoiling passivity. The small- K phase reproduces the optical/Gordon limit.

Proton-like core (heavy chain). A compact graph with large couplings $|t_{ij}|$ acts as a heavy, nearly static source; its field is the small- K face of $\Xi(K)$ (Newton/Coulomb scaling), while radiative pieces live in higher orders of K .

Hydrogen as a two-chain bound state. Let an electron-like surfeon chain move on S in the field of a proton-like core. Momentum-space shifts are linear functionals of $\Xi(K)$ with hydrogenic weights $W_{n\ell}(K)$:

$$\Delta E_{2S \rightarrow 2P} = C \int_0^\infty \Xi(K) [W_{2S} - W_{2P}] dK, \quad \Delta E_{1S} = C \int_0^\infty \Xi(K) W_{1S} dK. \quad (2)$$

Fix the single normalization C on a centroid (e.g., $2S \rightarrow 2P$). A *positive UV spectral density* (Compton threshold) in the Stieltjes representation of Ξ can supply the high- K suppression for 1S, without touching the IR constants.

5 The Spin-2 Graviton

Boundary law (TT sector). In transverse-traceless gauge, the linear SR relation reads schematically

$$[K_{ab} - K \gamma_{ab}]^{\text{TT}} = 8\pi G (Y^{(2)}(\omega, K) I + s^{(2)}(K) J) h_{ab}^{\text{TT}}, \quad Y^{(2)} \propto \Xi(K) \ (K \rightarrow 0), \quad (3)$$

so the universal spin-2 admittance is set by the same Ξ ; $s^{(2)}$ encodes a tiny parity-odd rotation that flips under $\hat{n}$ reversal.

Static limit and $1/r$. The static propagator induced by Ξ reproduces the massless spin-2 bulk Green function,

$$V(r) \propto \int \frac{d^3 q}{(2\pi)^3} \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{q^2} = \frac{1}{4\pi r}, \quad (4)$$

yielding Newton's $V(r) = -GM/r$ after matching to Cassini/Shapiro normalization. The Cassini corridor $\max_b |\gamma - 1| \leq 2.3 \times 10^{-5}$ constrains the small- K neighborhood of Ξ ; parity effects proportional to $s^{(2)}$ are correspondingly tiny but falsifiable (normal-flip signature).

6 Snapshots and Checks

GR optics/timing (IR window). The small- K expansion of Z and s reproduces weak-field lensing/time-delay with Cassini-level constraints. In the S-ring view, the long-wave limit fixes universal, channel-wise IR constants shared across sectors; surfeons inherit these as low-energy data.

Atomic integrals (QED overlays). Hydrogenic contrasts (e.g., $2S \rightarrow 2P$) probe $\Xi(K) := \Re Z(0, K)$ via momentum-space weights. A single normalization maps the shared IR kernel to the centroid; a UV *spectral density* (not a knee) can fold in Compton-scale screening for 1S without spoiling IR fits.

Gravitational-wave backgrounds. With IR constants frozen, S-ring edge templates predict shape-only anisotropy fractions and parity-odd residuals consistent with LVK O3 and NANOGrav corridors. The surfeon lens emphasizes edges and chirality; any detection of parity-odd backgrounds directly constrains $s(K)$.

7 Why This Framing Helps

- **Symmetry & economy:** particles as boundary automodes reduce to a two-slot algebra with positivity constraints; few IR constants propagate across domains.
- **Causality baked in:** spectral measures guarantee Herglotz/Stieltjes structure (no ad-hoc knees), keeping the surfeon causal and passive by construction.
- **Interoperability:** the algebra and star-product describe how surfeons compose and couple to bulk or other surfaces, matching modern scattering/network viewpoints.
- **Falsifiability:** parity flips (orientation), anisotropy shapes, and cross-domain ratios offer parameter-light tests once the IR is fixed.

8 Attribution, Feedback, and Companion Materials

How to cite. “Surface Rings (S-rings) as Particles (Front-End)”, H. Salval, November 6, 2025, v0.1. Companion: “Surface Rings: Formal Definitions, Characterization, and Cross-Domain Validations” (proofs, Dirichlet forms, boundary triples, canonical examples).

Companion files (repository). Formal note: `Research Journal/SR.Formalization_and_Validations.tex`. Legacy front paper (context): `Research Journal/Surface_Rings_front_paper.tex`.

Feedback welcome. This is a live front-end intended for community review. Please share comments on the repository issue tracker or by sending notes referencing specific equations or sections. Priority topics:

- independent checks of the IR calibration (Cassini corridor, multi- b tracks);
- measure-based recomputation of hydrogen integrals (centroids and 1S scaling);
- GW anisotropy shape matches and parity-odd bounds.

Note on numerics. Earlier overlays used knee fits; all results are being recomputed with *positive spectral measures*. Qualitative conclusions persist; quantitative values will be updated as the new fits converge.

References (selected pointers)

- Schilling, Song, Vondraček, *Bernstein Functions* (de Gruyter) — Stieltjes/Bernstein classes and subordination.
- Fukushima, Oshima, Takeda, *Dirichlet Forms and Symmetric Markov Processes* (de Gruyter) — Energy forms and induced geometry.
- Grubb, *Distributions and Operators* (Springer); McLean, *Strongly Elliptic Systems and Boundary Integral Equations* — DtN/Calderón operators.
- Jacob, Zwart, *Linear Port-Hamiltonian Systems on Infinite-Dimensional Spaces* (Springer) — Positive-real/impedance viewpoint.
- Nevanlinna, *Analytic Functions*; Widder, *The Laplace Transform* — Herglotz and Stieltjes functions.
- Gordon (optical metric): Plebanski (1960), Leonhardt (2006) for effective index/gravity optics.